

				
Ground Truth	woman hold knife knife cut meat meat placed on chopping board chopping board on table	hand hold towel water wet towel tower scrub baby baby hold toy duck basin filled with water toy duck float on water	left hand hold handcuffs right hand hold woman handcuffs handcuff woman	syringe inserted into slice of bread wheat next to slice of bread
ClipCap	man hold knife knife cut meat meat placed on chopping board chopping board on table	right hand hold toothbrush toothbrush inside mouth	crowd sit on chair crowd look at man	left hand press bread right hand hold knife knife cut bread
Git	man hold hammer hammer beat nail nail nailed to wooden board	boy hold toy duck toy duck in bathtub	crowd sit on chair crowd look at man	electric drill drill bread
Ours	man hold knife knife cut meat meat placed on chopping board chopping board on table	boy sit in bathtub boy hold toy toy immersed in water water in bathtub	man hold handcuffs handcuffs handcuff woman	syringe pierce bread bread placed on chopping board

Figure 7: Comparisons of triplets generation across diverse OVRE methods. The illustration highlights accurately described triplets in green, triplets with semantic correlation in blue, and irrelevant triplets in red.

cally, we utilize RegionCLIP (?) pre-trained from LVIS to crop bounding boxes and seqNMS (?) for object tracking. (II) Frame features: We directly utilize features extracted from individual frames using CLIP, concatenating them to form a representation of frame-level features. As depicted in Table 4, both frame features and region features exhibit poor performance. Notably, frame features capture the overall visual content of an image but overlook finer details such as objects and relationships. Surprisingly, region features fare even worse compared to frame features. We hypothesize that this is attributed to the limited generalization capability of existing object detectors. The diverse range of object categories complicates their accurate detection within our Moments-OVRE context.

Conclusion

In this paper, we introduce a new task named OVRE, where the model is required to generate all relationship triplets associated with the video actions. Concurrently, we present the corresponding Moments-OVRE dataset, which encompasses a diverse set of videos along with annotated relationships. We conduct extensive experiments on Moments-OVRE and demonstrated the superiority of our proposed approach over other baseline methods. We hope that our task and dataset will inspire more intricate and generalizable research in the realm of video understanding.

Limitations: (I) This version of Moment-OVRE has currently omitted BBox annotation due to the high cost of annotation. We are committed to progressively enhancing this dataset and intend to introduce BBox annotations in upcoming

versions of Moments-OVRE. (II) For extracting action-centric relations, leveraging commonsense among action categories and relations (?) or implicit knowledge-driven representation learning methods (??) have shown promise. We will consider these knowledge-driven methods in future work.

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